



RACK-SCALE ARCHITECTURE
FOR WAFER SCALE ENGINE

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Co-Founder & Chief System Architect

Cerebras Systems

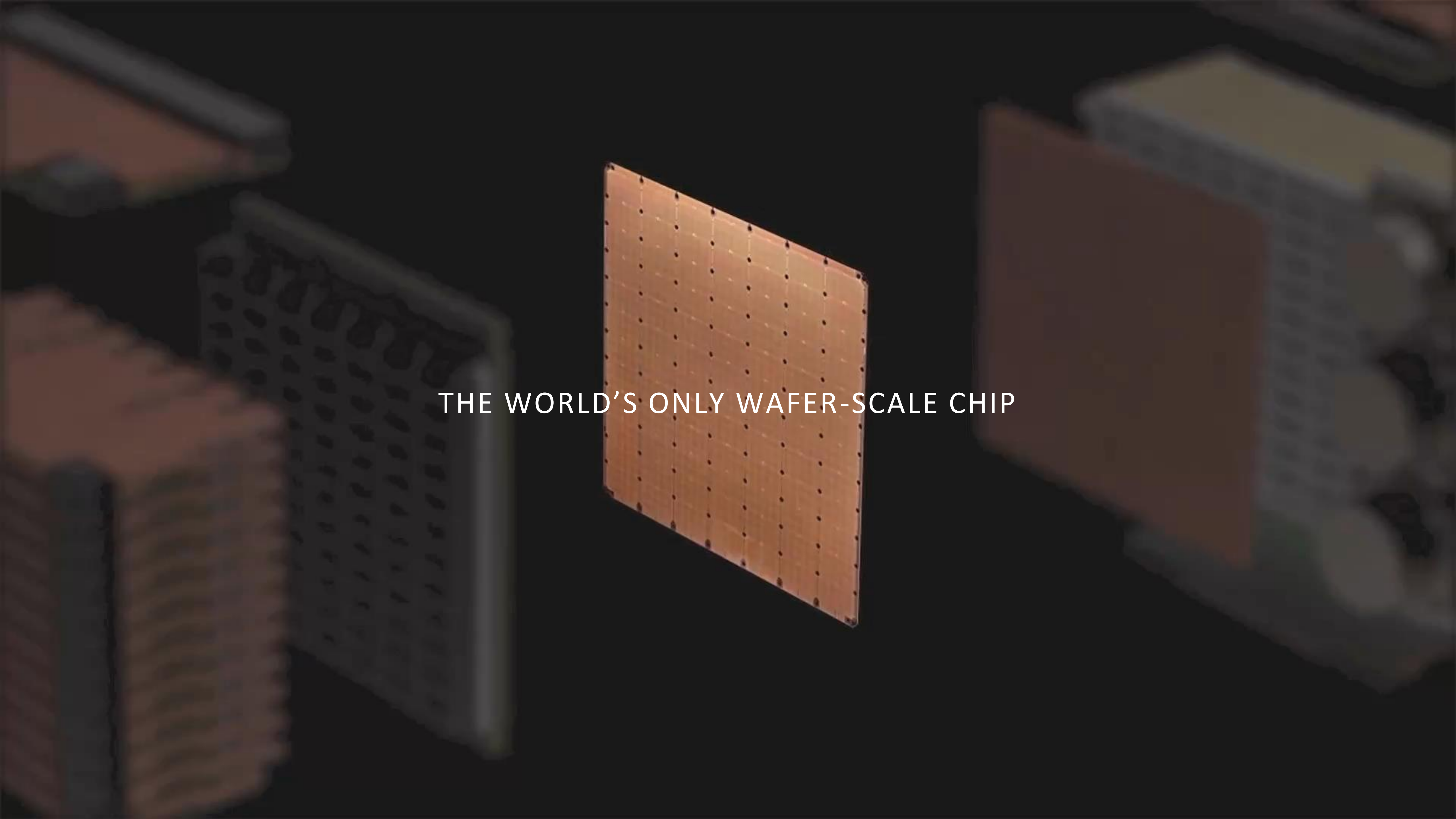
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A copper wafer-scale chip with a grid of small holes, surrounded by other electronic components in a dark environment.

THE WORLD'S ONLY WAFER-SCALE CHIP



INTRODUCING

THE CEREBRAS CS-4

THE CEREBRAS CS-4

UP TO

2x

FASTER TOKENS

UP TO

10x

THROUGHPUT / WATT

VS. Industry-Leading CS-3

CS-4 Speed Advantage



INFERENCE ON CEREBRAS IN PRODUCTION *TODAY*

Output Speed
(Tokens/sec)

● GPU

● CS-3

5,000

4,000

3,000

2,000

1,000

0

0

Gemma
4-31B

Llama
3.3 70B

GPT
OSS-120B

GLM
4.7-355B

Kimi
K2.7 1T

GPT
5.4

GPT
5.6 Sol

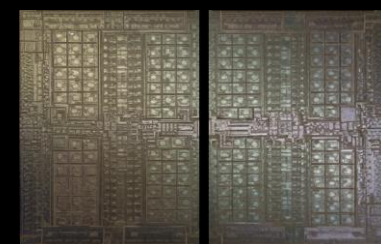
Source: Artificial Analysis and internal benchmarking (August 2026)

CS-4 IS TRANSFORMATIVE: *UP TO 30X FASTER*

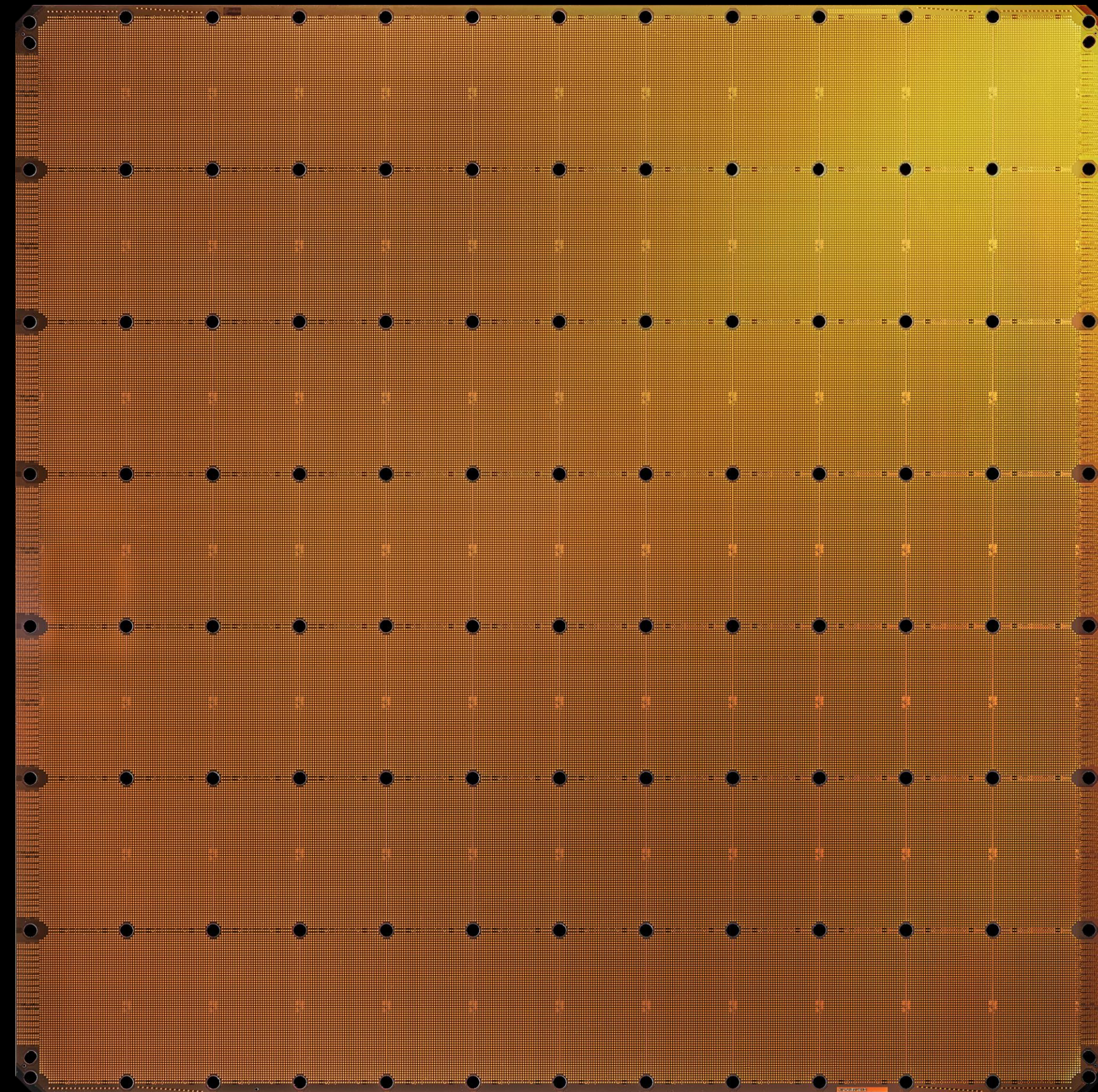


Source: Artificial Analysis and internal benchmarking (August 2026)

2,000x MORE MEMORY BANDWIDTH THAN RUBIN

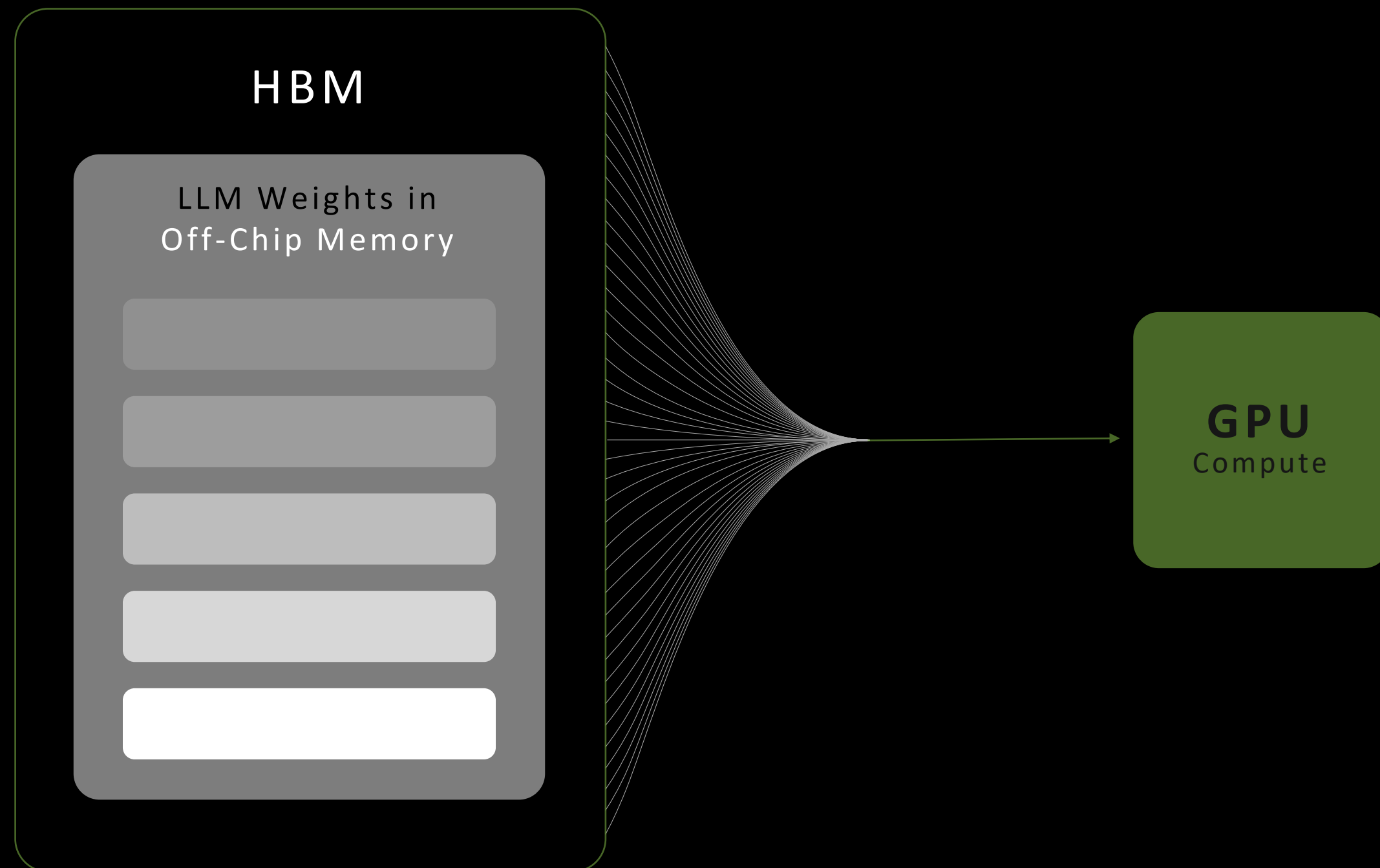


NVIDIA Rubin
22 TByte/s



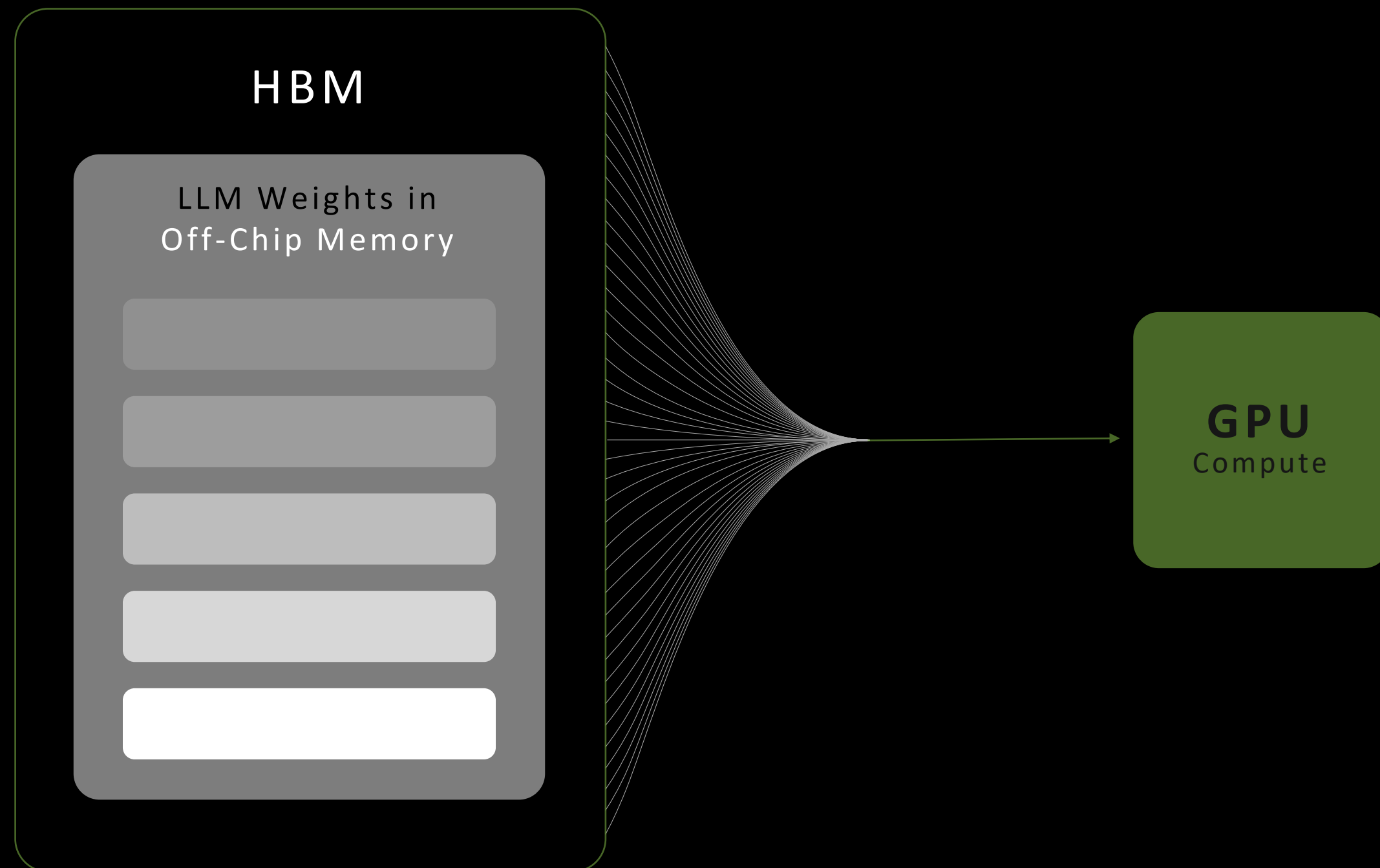
Cerebras CS-4 WSE-3T
43,200 TByte/s

GPU



NVIDIA Rubin
22 TByte/s

GPU



NVIDIA Rubin
22 TByte/s

CEREBRAS WSE-3T



Cerebras CS-4 WSE-3T
43,200 TByte/s

NEXUS RACK-SCALE PLATFORM

DESIGNED FOR MODULARITY

- 01 Simpler to build
- 02 Faster to deploy
- 03 Independent innovation of power, compute, IO

POWER DELIVERY IN THE
FRONT OF THE RACK



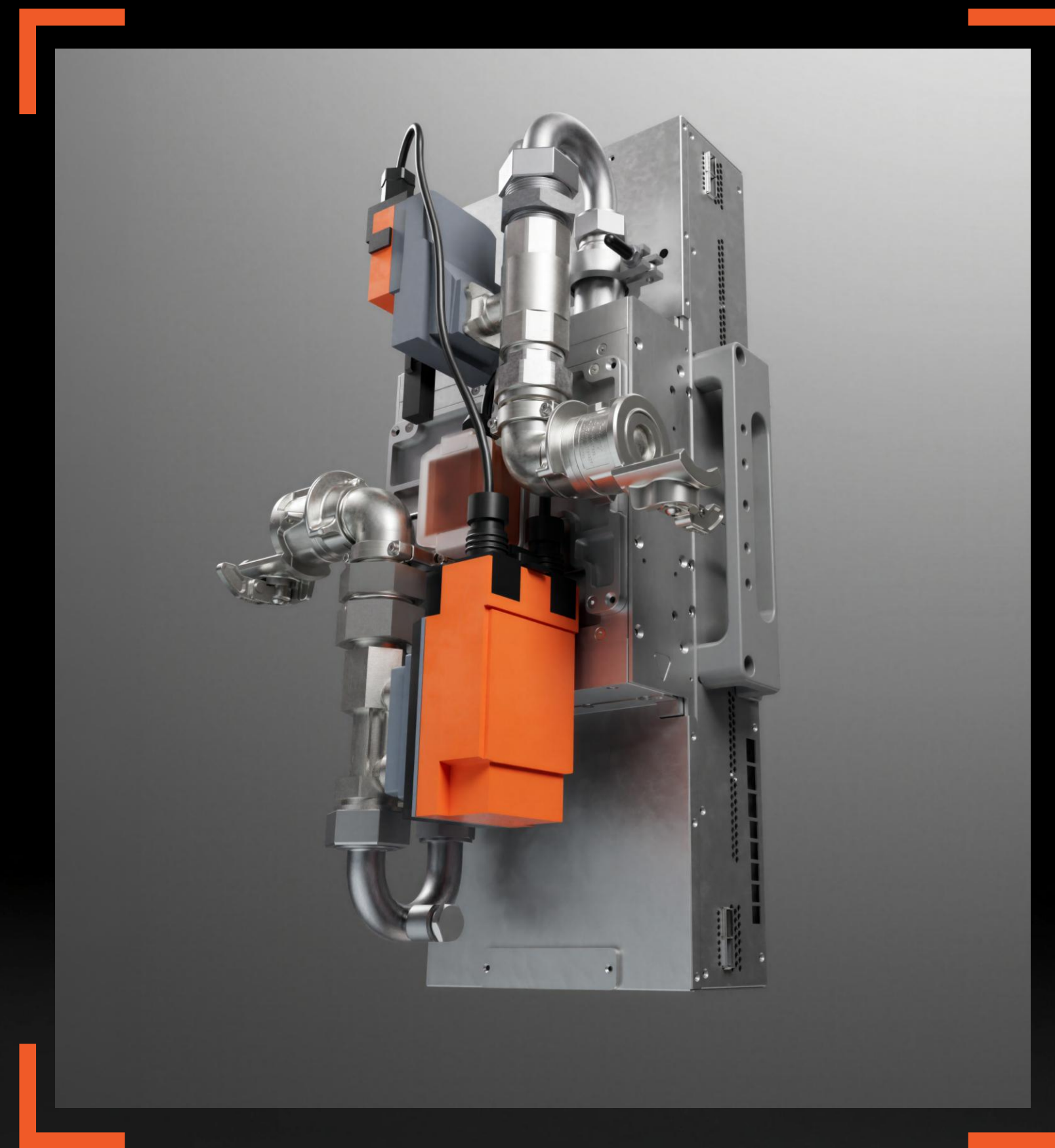
COMPUTE IN THE REAR IN
PLUGGABLE BACKPACKS

BACKPACK CONTAINS
WAFER SCALE ENGINE

All comparisons relative to Cerebras CS-3



3 WAFER-SCALE
BACKPACKS
PER CS-4



All comparisons relative to Cerebras CS-3

CS-4 BACKPACK:
NEXT-GEN PERFORMANCE

WAFER IO MODULE

2x more bandwidth
2x faster latency
Modular upgrades

WAFER PACKAGE

Direct vertical power
2x more power
2x more cooling

EFFICIENT MANUFACTURING

50% fewer components
60% more automation

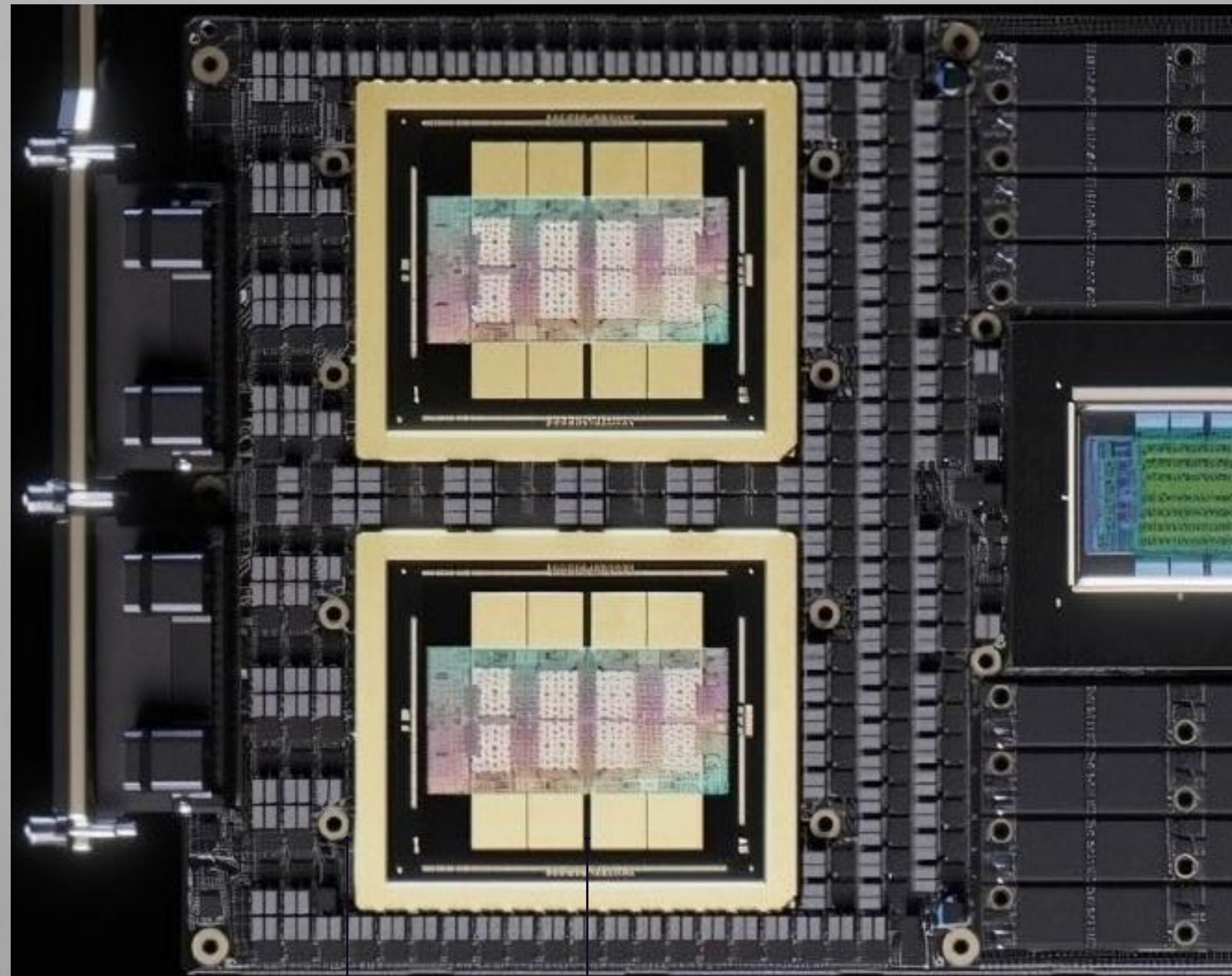
All comparisons relative to Cerebras CS-3

INEFFICIENT POWER DISTRIBUTION OF GPU

~50MM BETWEEN POWER CONVERTERS
AND SILICON LOAD

Power losses along path inside
Printed Circuit Board

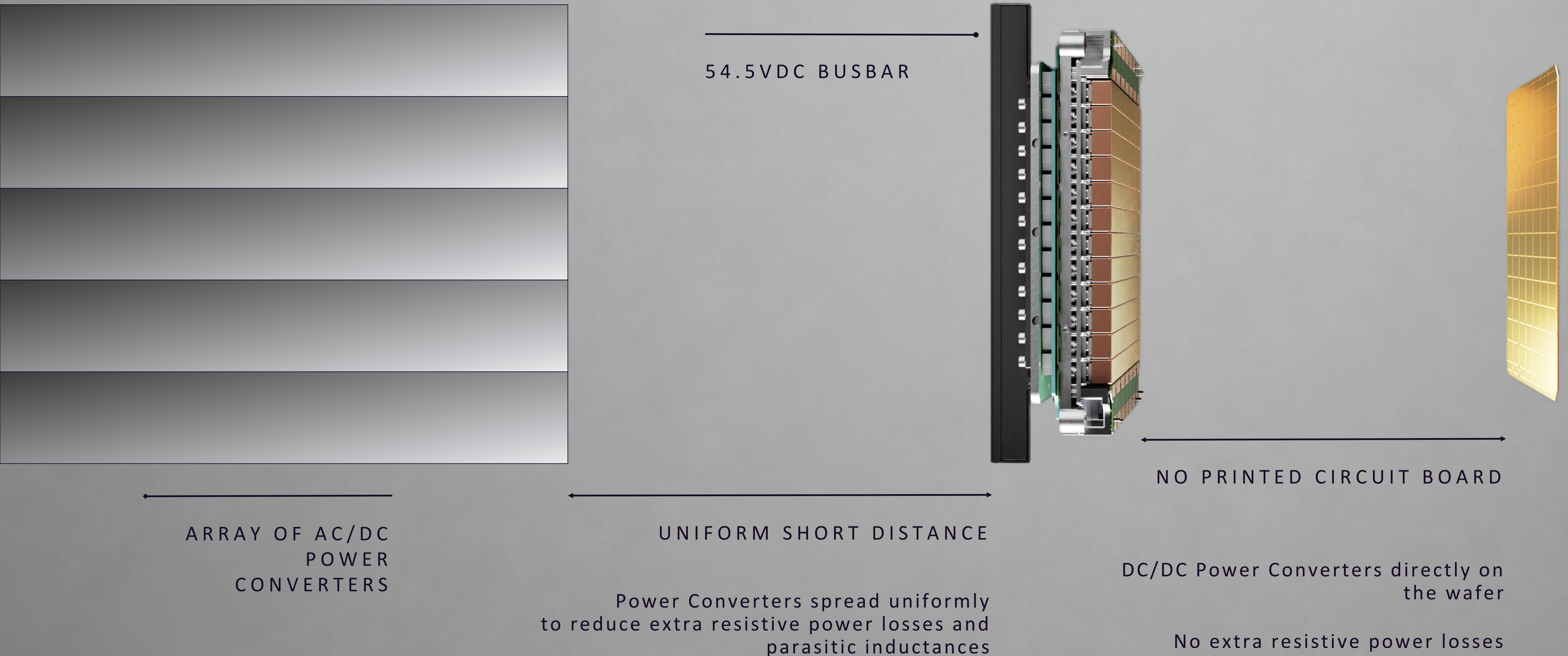
Requires more copper layers to
reduce resistive losses — extra
cost & complexity



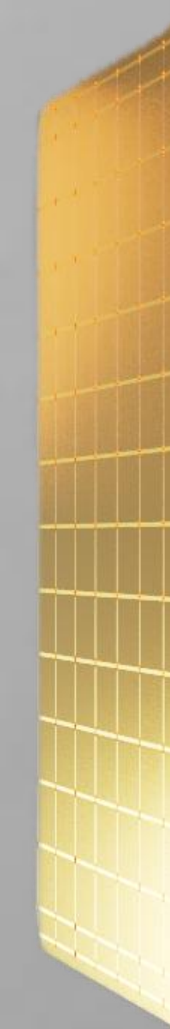
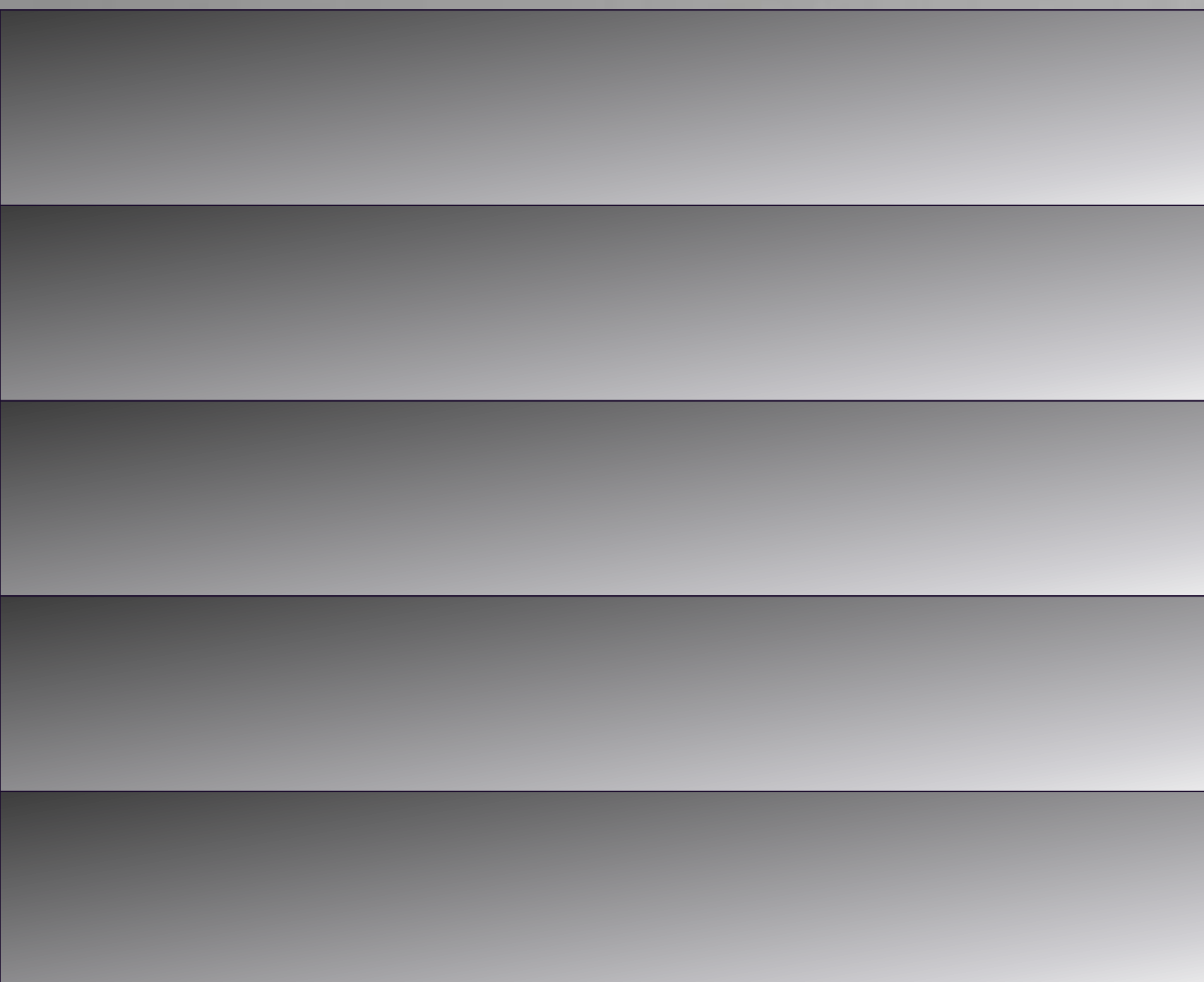
~50mm

Image Source: nvidia.com/en-us/data-center/vera-rubin-nvl72

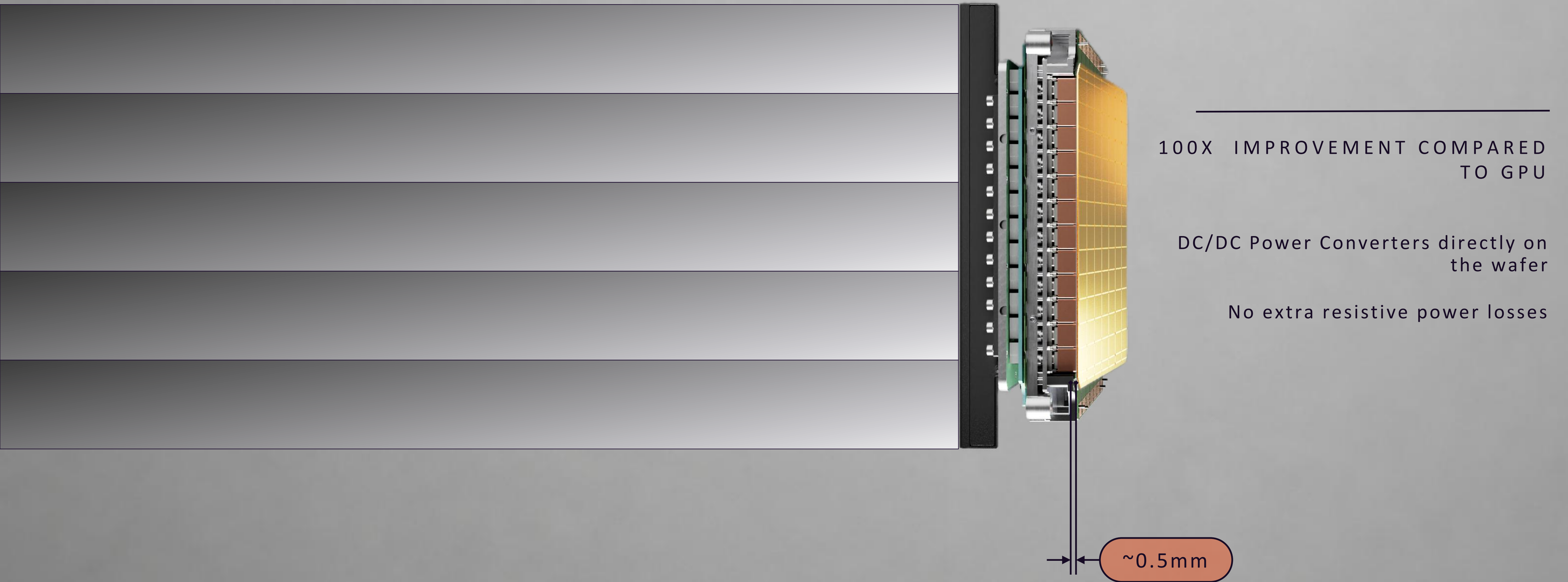
EFFICIENT POWER DISTRIBUTION



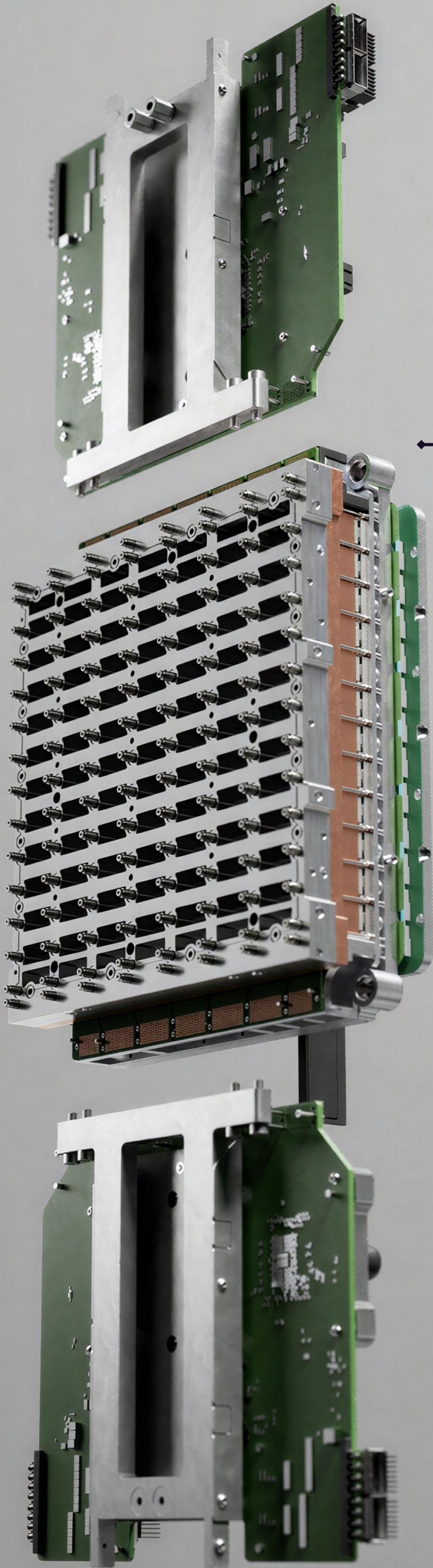
EFFICIENT POWER DISTRIBUTION



EFFICIENT POWER DISTRIBUTION



THE NEXT GEN
WAFER IO INTERFACE



HIGHER BANDWIDTH
AND LOWER LATENCY

New direct wafer link interfaces
Standard RoCE protocol network

NEW WAFER IO MODULE

Extends fabric from wafer edges
Modular & programmable for the future

INTEGRATED WATER CONDITIONING

FLOW REGULATION ACTUATOR

Guarantees proper water flow rate

LEAK DETECTION MODULE

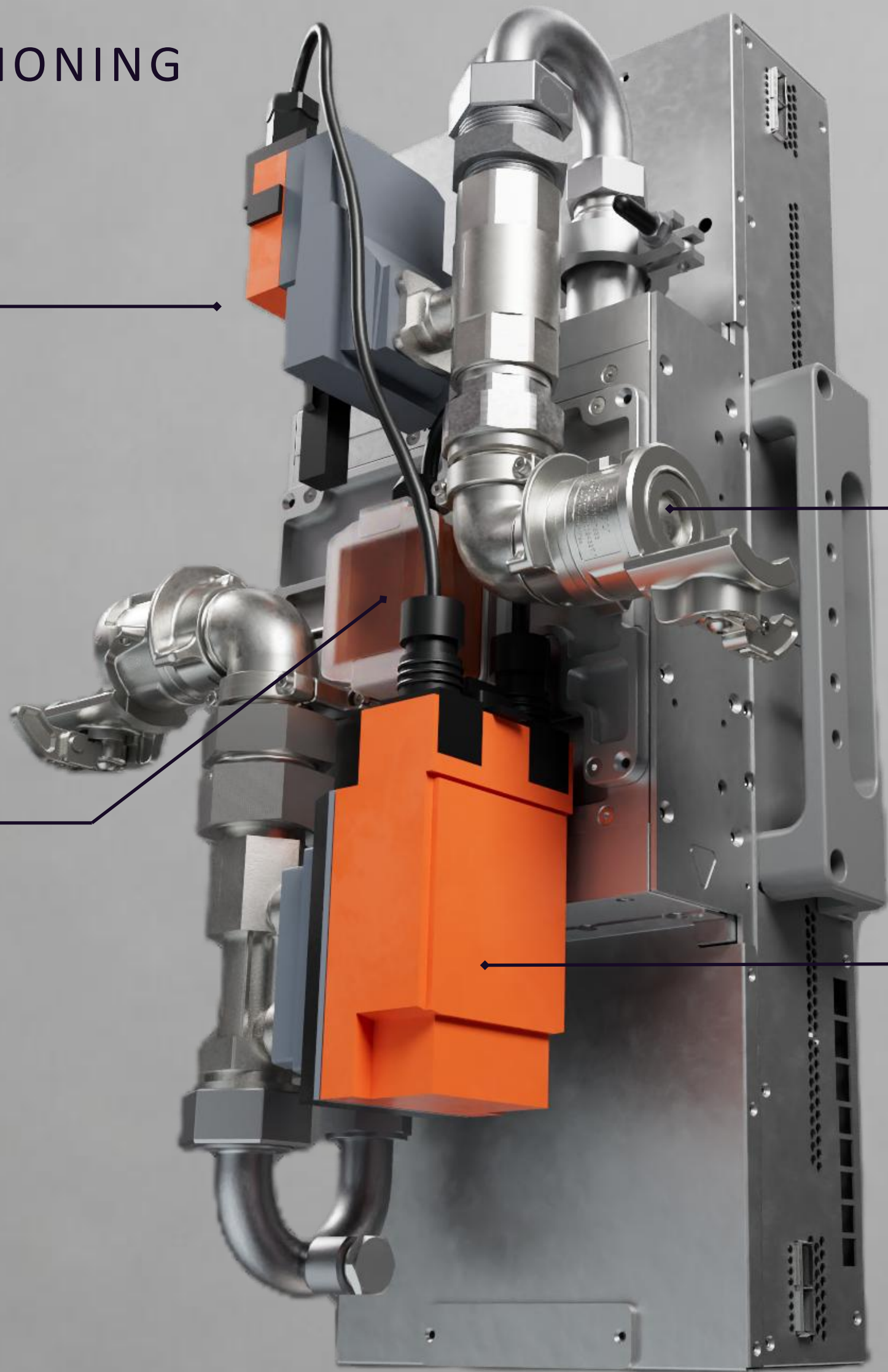
Monitors leaks and condensation
Shuts off AC power to all the
Power Supply modules

VALVED DRY QUICK
DISCONNECTS

For easy install

ENERGY METER

Monitors water flow rate
Monitors water inlet & outlet temp.



WATER & COMPUTE IN THE REAR

INTEGRATED WATER DELIVERY

Supply and return manifolds on each side
Valved Dry Quick Disconnects for easy install
isolated from high-voltage AC power section in the front

CS-4 BACKPACK

Backpack can easily be swapped

INTEGRATED FIBER CONDUIT

Fiber optic cables are protected



POWER IN THE FRONT

AC/DC POWER SUPPLY MODULES

Input: up to 277VAC, Output 54.5VDC

Up to 30x modules per backpack

Flexible power feed redundancy:
5+1, 4+1, 3+1, 4+2

Air-cooled

FAN MODULES

Cool small devices in the Backpack



INTEGRATED POWER PROTECTION

Each Power Supply is protected by
a 30A circuit breaker behind cover
panels

POWER IN THE FRONT

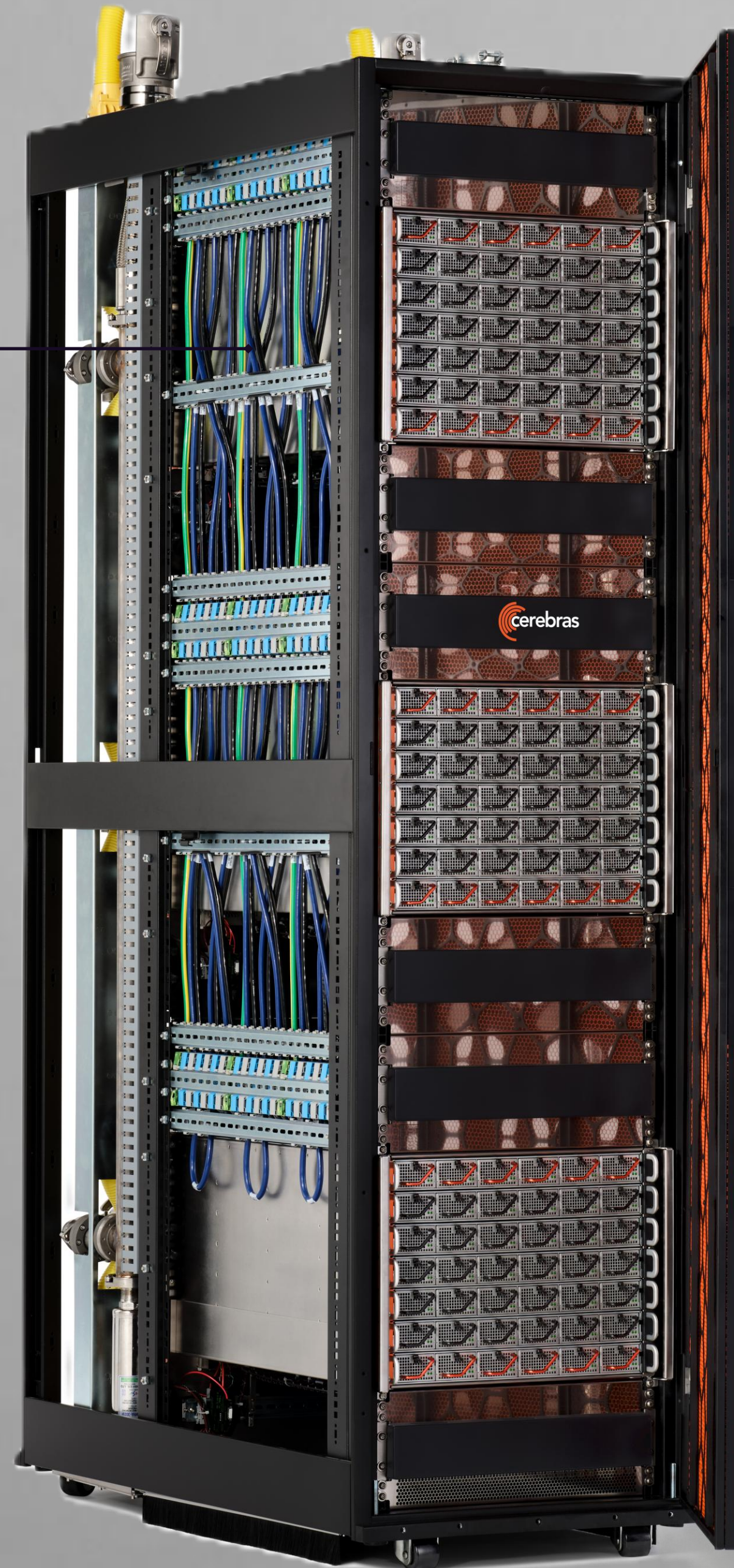


POWER IN THE FRONT

INTEGRATED POWER DELIVERY

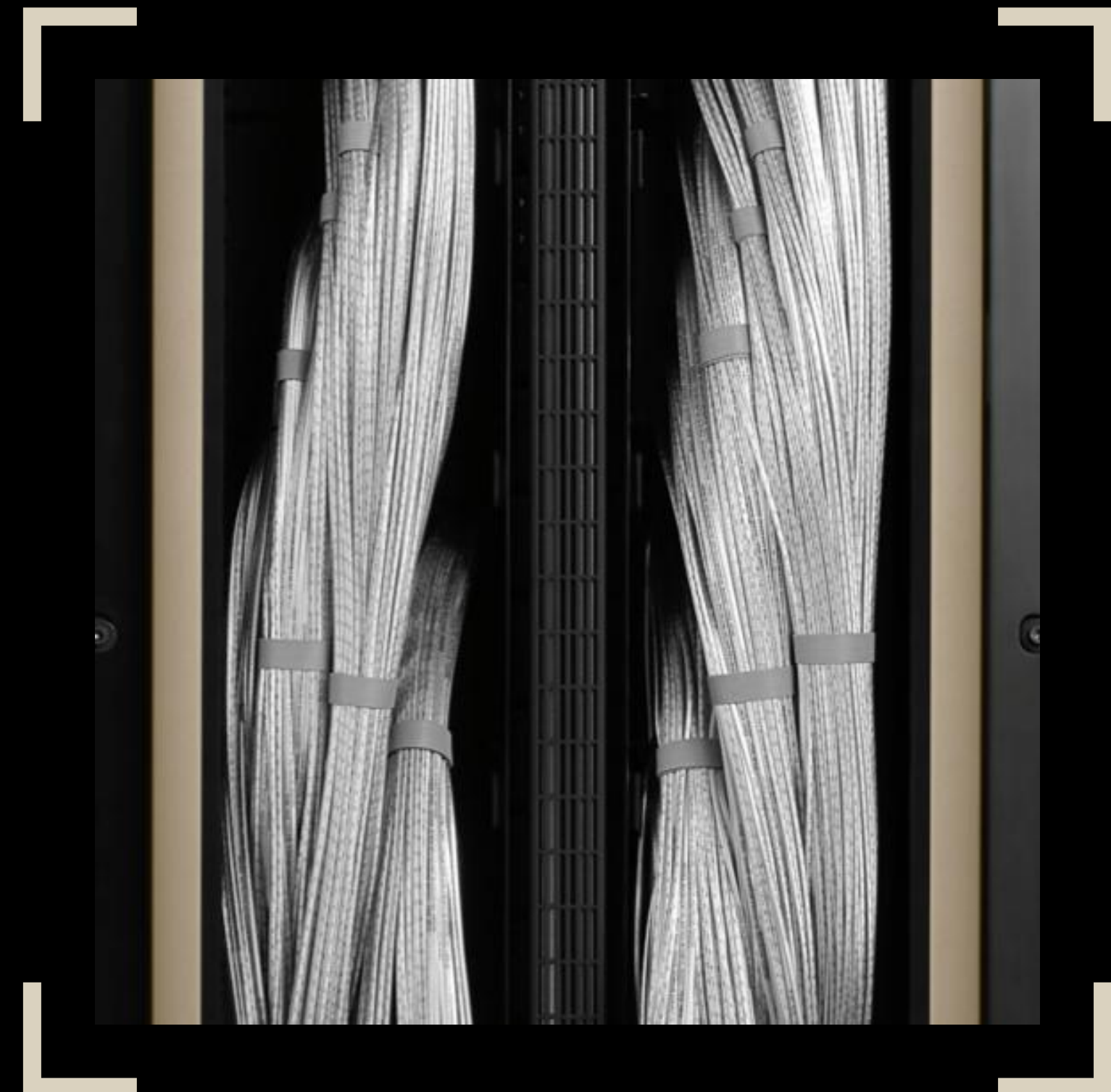
Power is fed by up to 6 AC feeds from the top of the rack

Fully phase balanced AC power delivery interconnect inside both sides of the rack

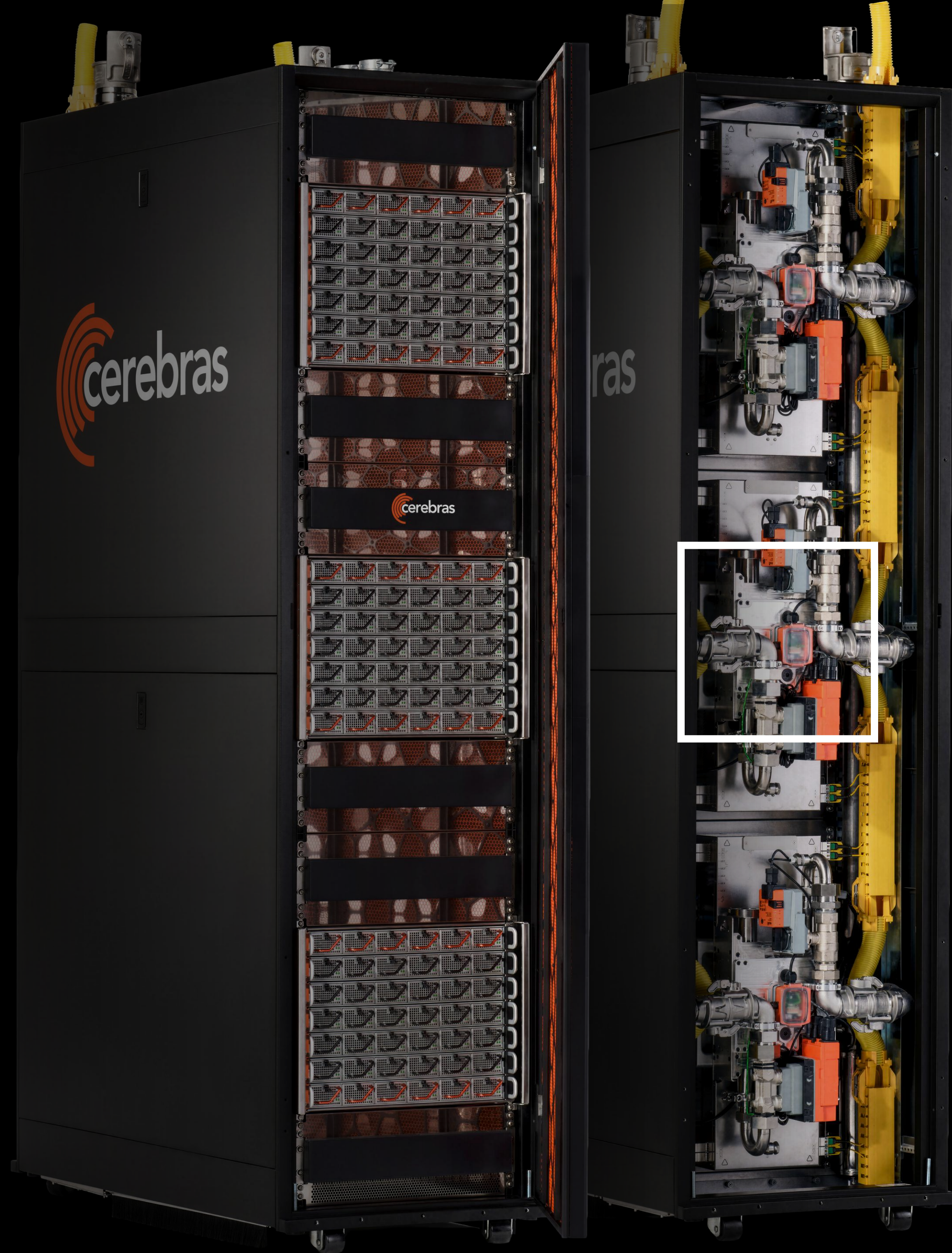




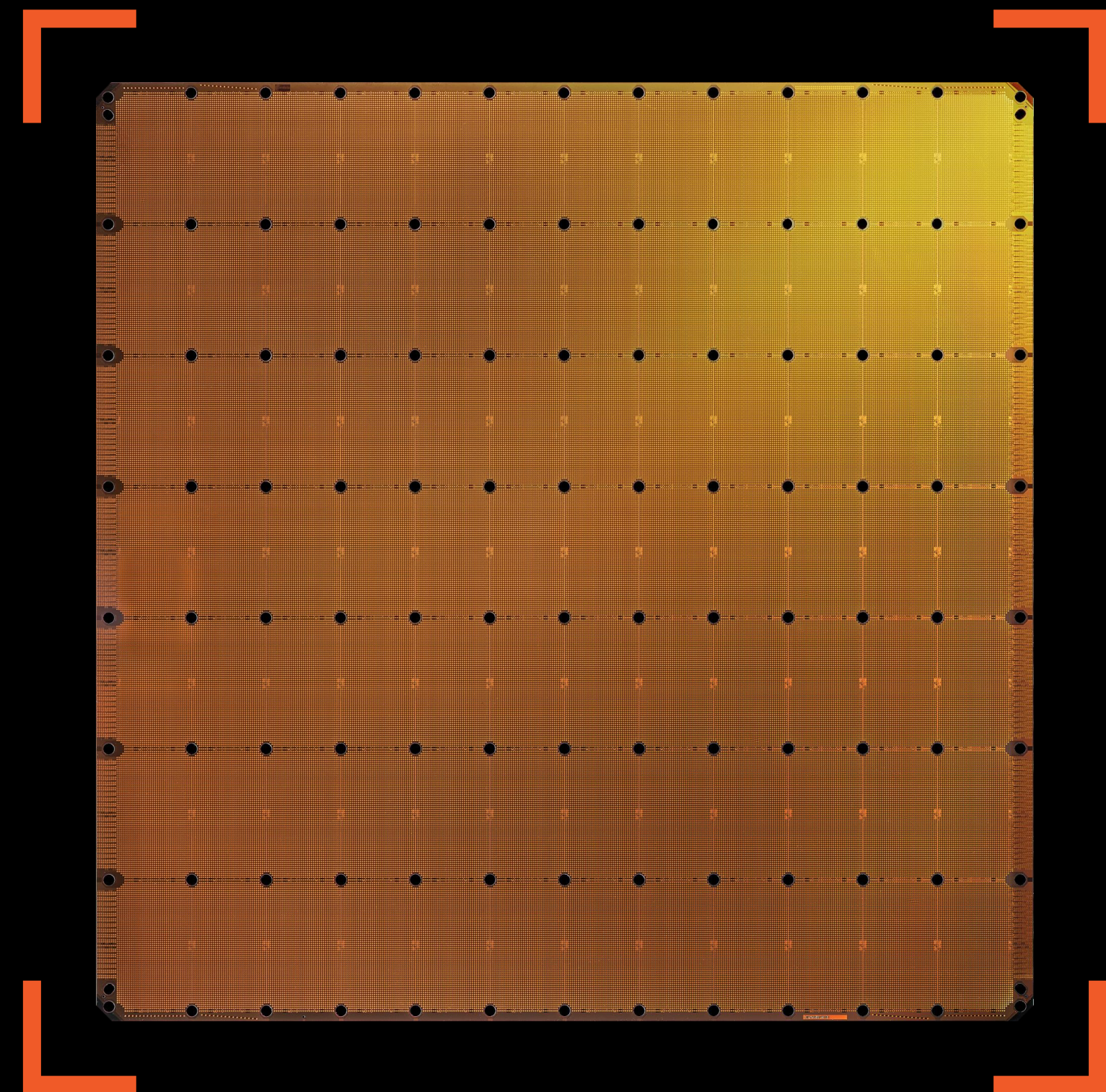
RUBIN NVL72
260 TB/S NVLINK
5,000 CABLES



“MORE BANDWIDTH
THAN THE INTERNET”



CS-4 WSE-3T
53.5 PB/S WAFER FABRIC
NO CABLES, ALL ON-CHIP

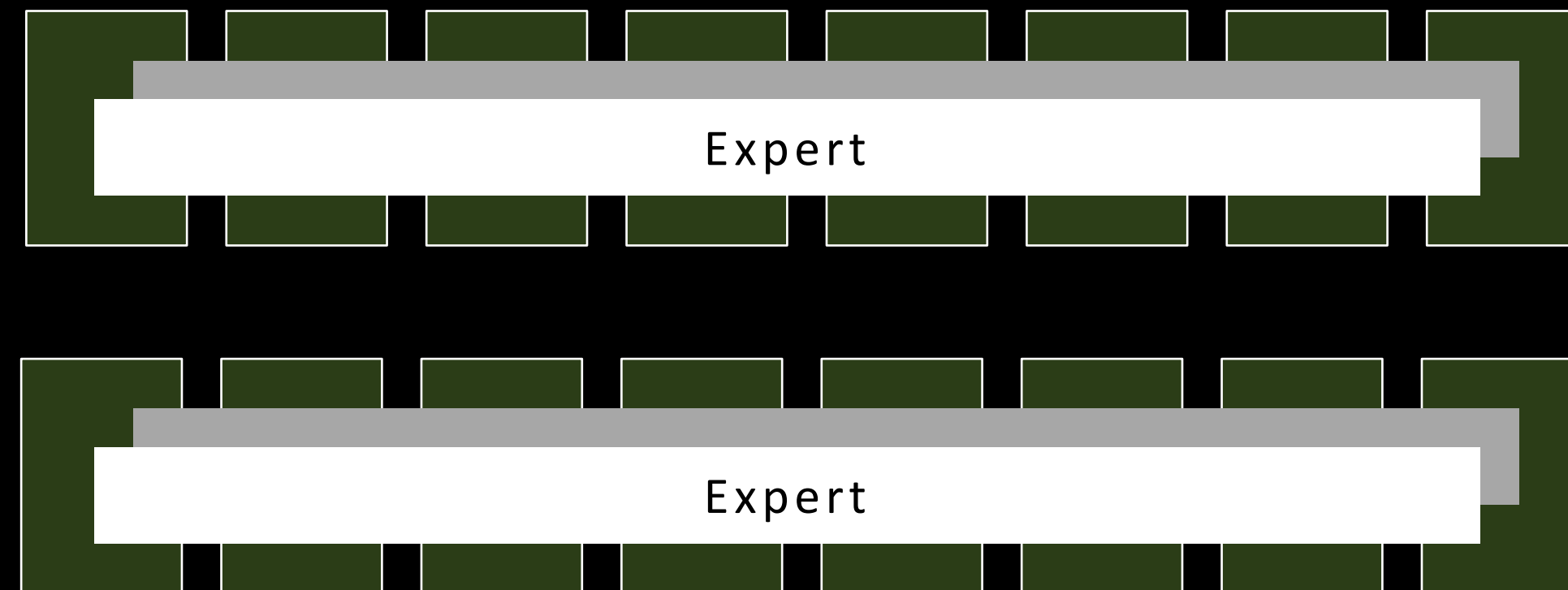


200X HIGHER BANDWIDTH THAN GPU
INTERCONNECT

CS-4 BY THE NUMBERS

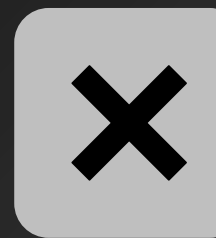
	CS-3	CS-4
AI Compute	125 PFLOPS	750 PFLOPS
Wafers	1x WSE-3	3x WSE-3 Turbo
Memory capacity	44 GByte	132 GByte
Memory bandwidth	21.6 PByte/s	129.6 PByte/s
Fabric bandwidth	26.7 PByte/s	160.5 PByte/s
IO bandwidth	1.2 Tbit/s	7.2 Tbit/s
IO latency	5 us	2 us

GPU

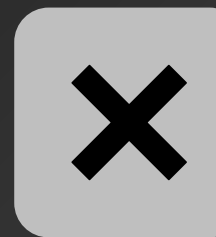


SLOWER.
HIGHER POWER.
HIGHER COST.

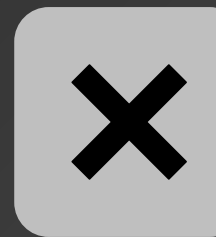
TRY TO WORKAROUND MEMORY BANDWIDTH



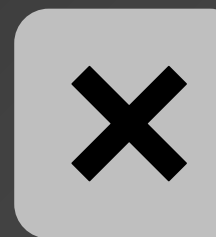
Experts distributed
across multiple GPUs



Tensor and expert
parallelism required

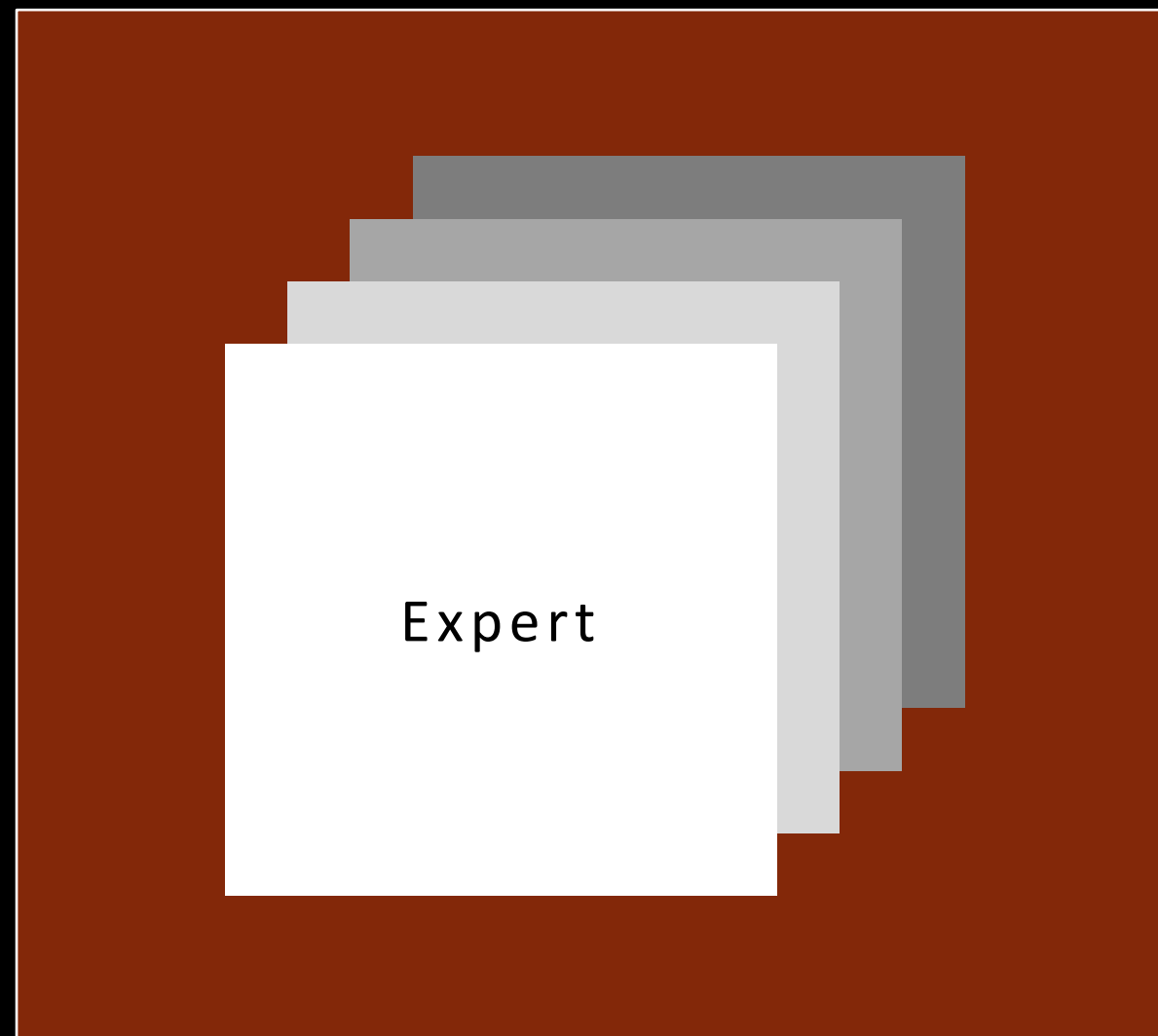


Attempt to aggregate
memory bandwidth



Complex communication
between GPUs

CEREBRAS WSE-3T



ULTRAFAST. SIMPLE.
EFFICIENT.

MEMORY BANDWIDTH ENABLES SPEED



All experts run on a
single chip



Experts interleaved on
wafer memory



No cross-chip
communication



No complex routing

HIGHER BANDWIDTH
LOWER LATENCY WITHIN
EACH WAFER

2X

BANDWIDTH PER WAFER

- ✦ New higher speed wafer links
- ✦ 2.4Tb/s aggregate bandwidth

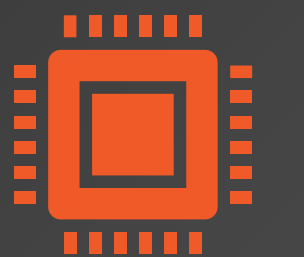
3us latency

CS-4 CLUSTER

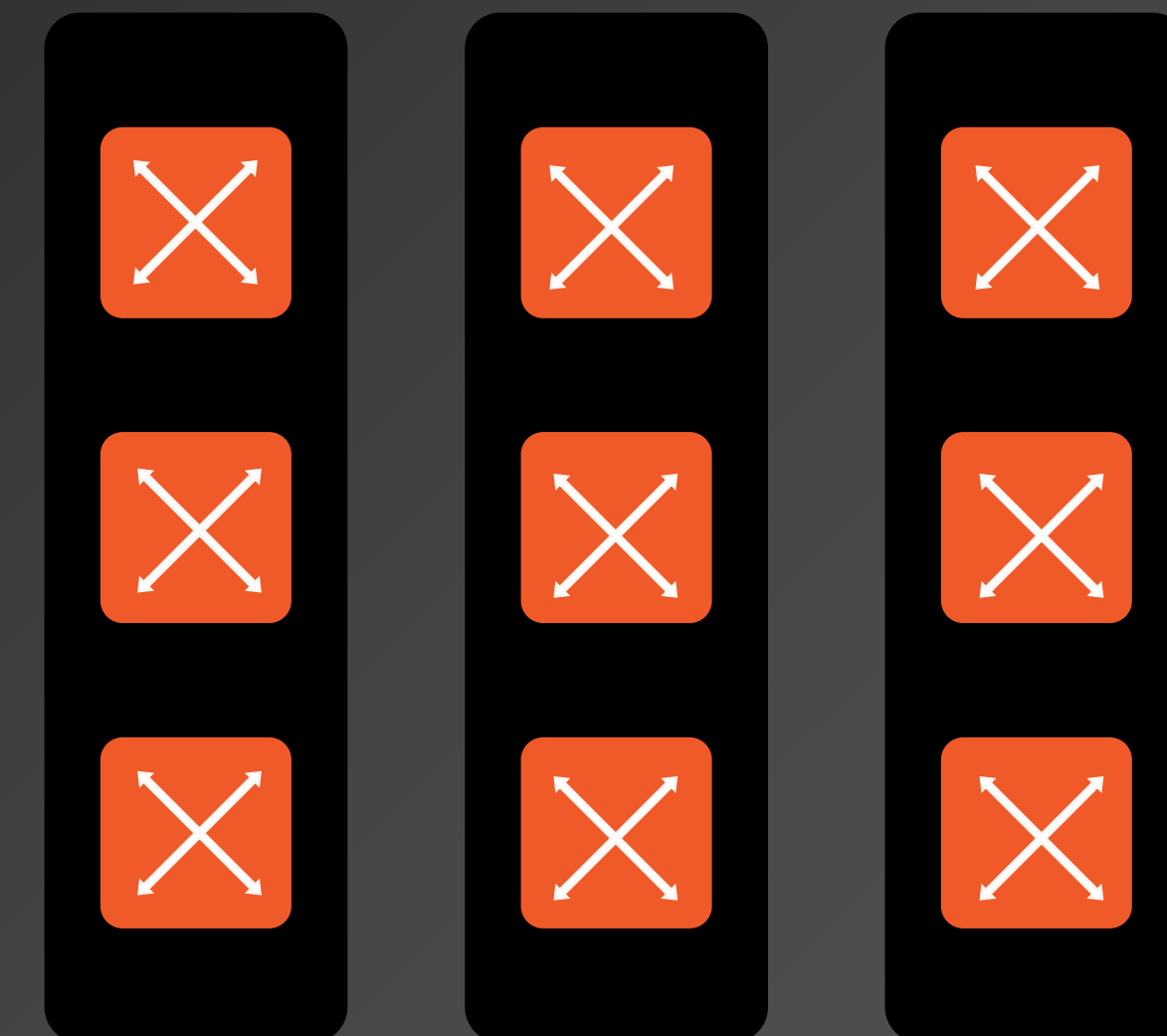
NETWORK



USERS



GPU / ASIC



HIGHER BANDWIDTH LOWER LATENCY BETWEEN WAFERS

2X

BANDWIDTH PER WAFER

- ✦ New higher speed wafer links
- ✦ 2.4Tb/s aggregate bandwidth

1.7X

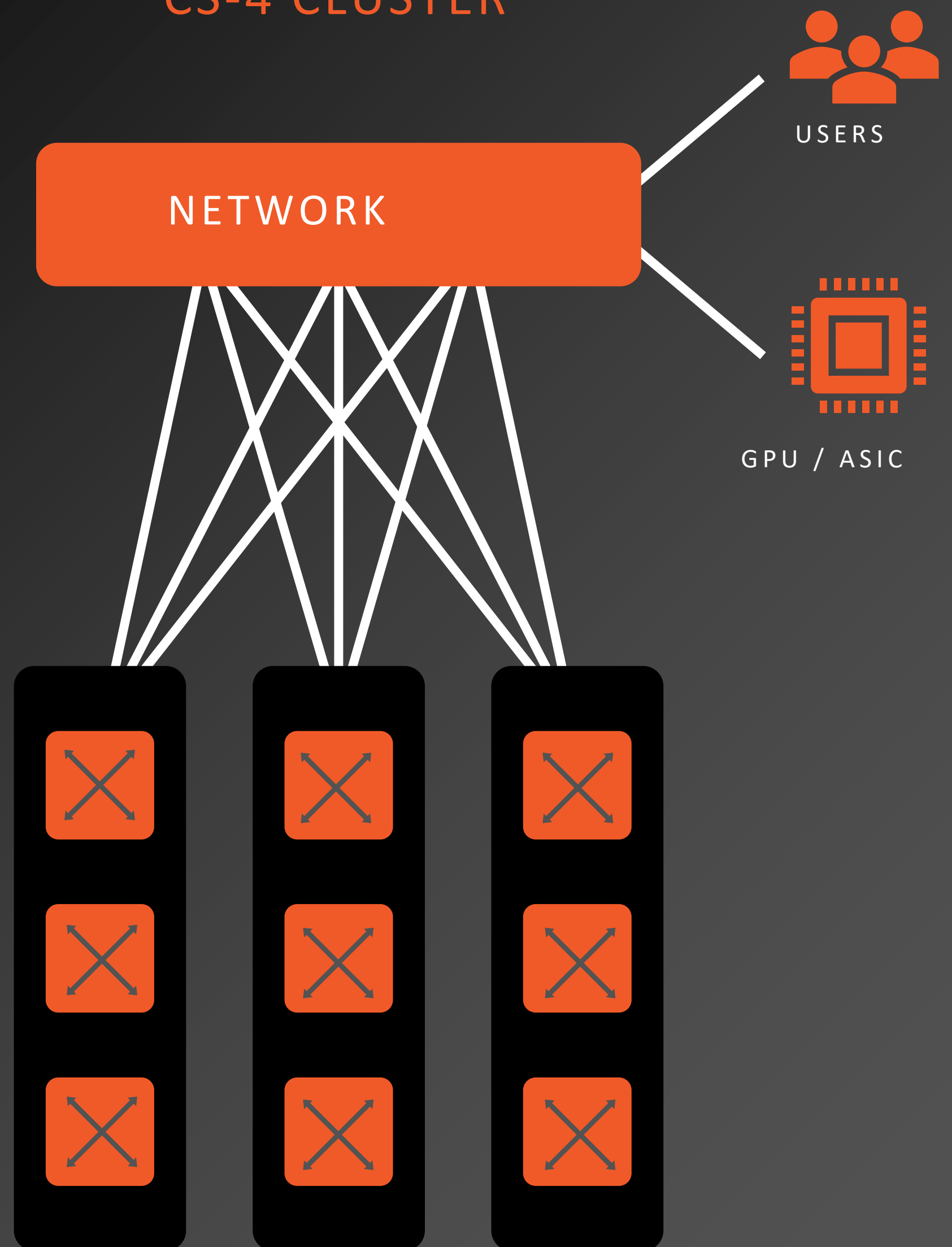
FASTER NETWORK LATENCY

- ✦ New low latency packet pipeline
- ✦ 3us latency to users & disaggregated devices

3us latency

2.4 Tb/s per wafer

CS-4 CLUSTER



HIGHER BANDWIDTH LOWER LATENCY BETWEEN WAFERS

2X

BANDWIDTH PER WAFER

- ✦ New higher speed wafer links
- ✦ 2.4Tb/s aggregate bandwidth

1.7X

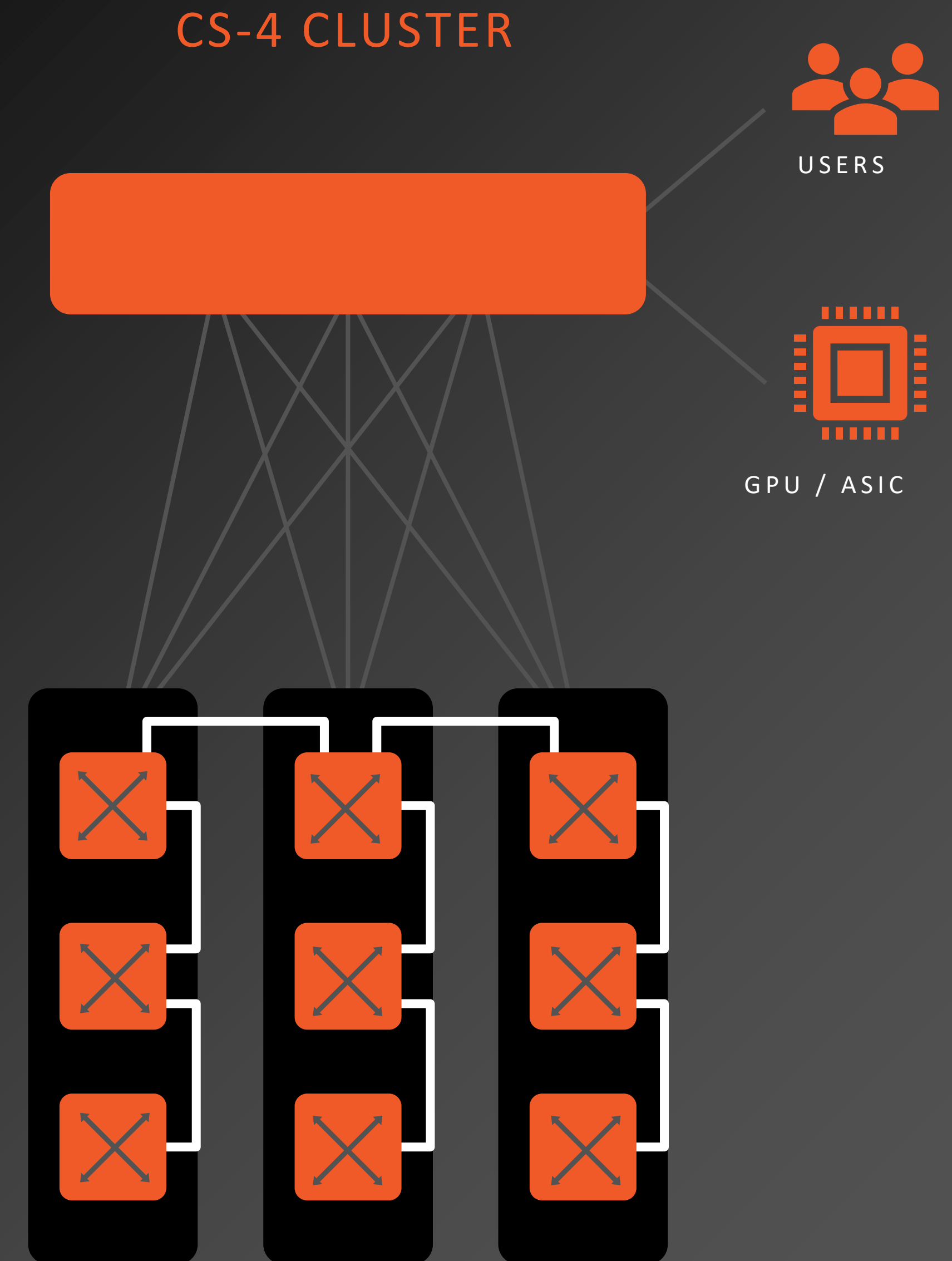
FASTER NETWORK LATENCY

- ✦ New low latency packet pipeline
- ✦ 3us latency through network

2.5X

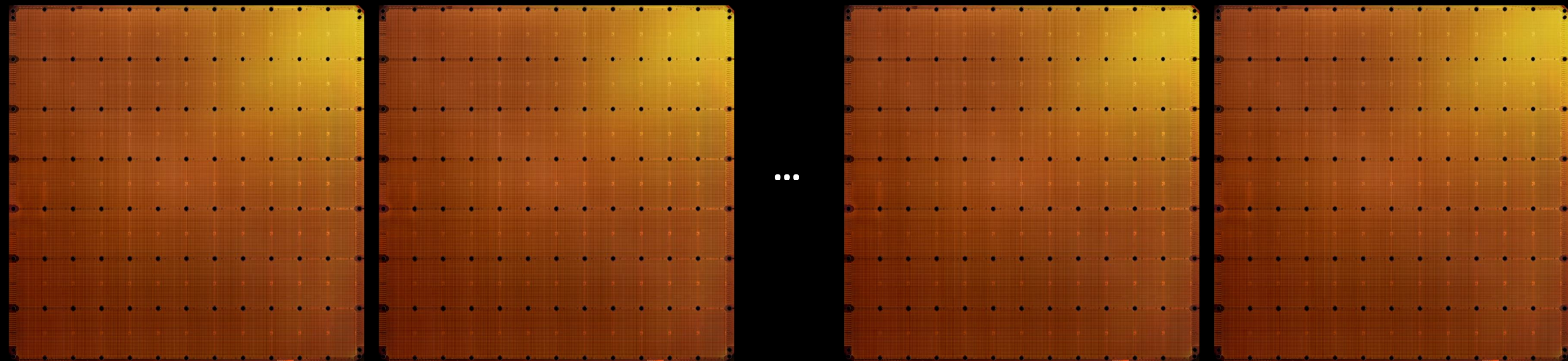
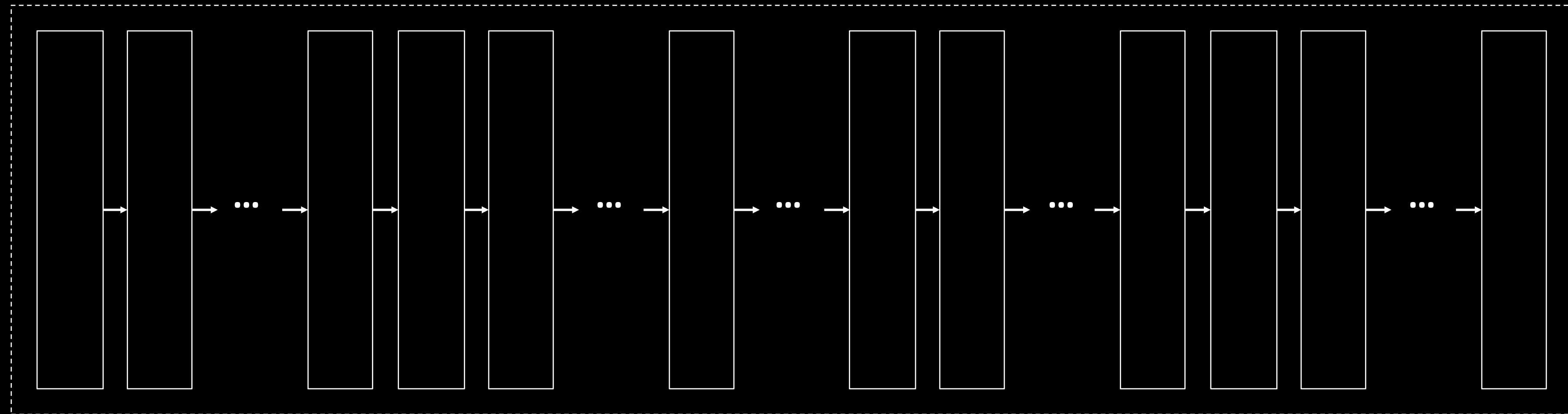
FASTER WAFER-TO-WAFER

- ✦ New Direct Wafer Links
- ✦ 2us latency wafer-to-wafer



CS-3 ALREADY RUNS THE LARGEST FRONTIER MODELS FAST

GPT-5.6 SOL



CS-3 CLUSTER

CS-3 ALREADY RUNS THE LARGEST FRONTIER MODELS FAST

01

Pipelining large models
across wafers is seamless
and fast

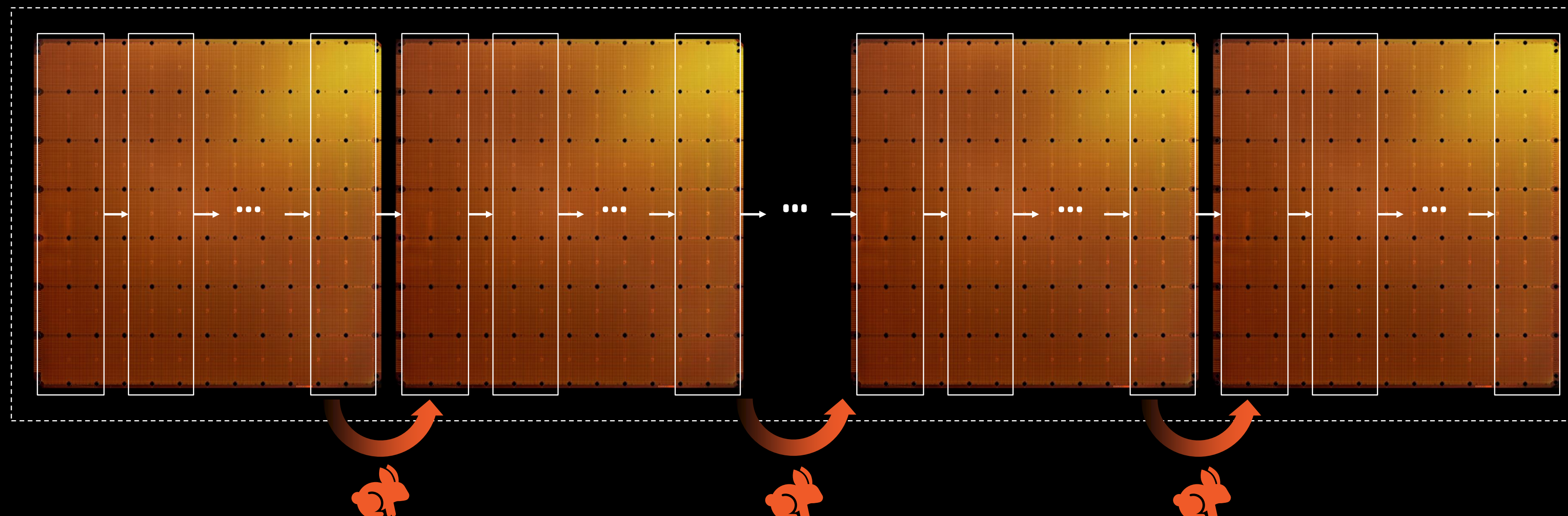
02

Keep almost all high communication
on wafer, using internal high
bandwidth fabric

03

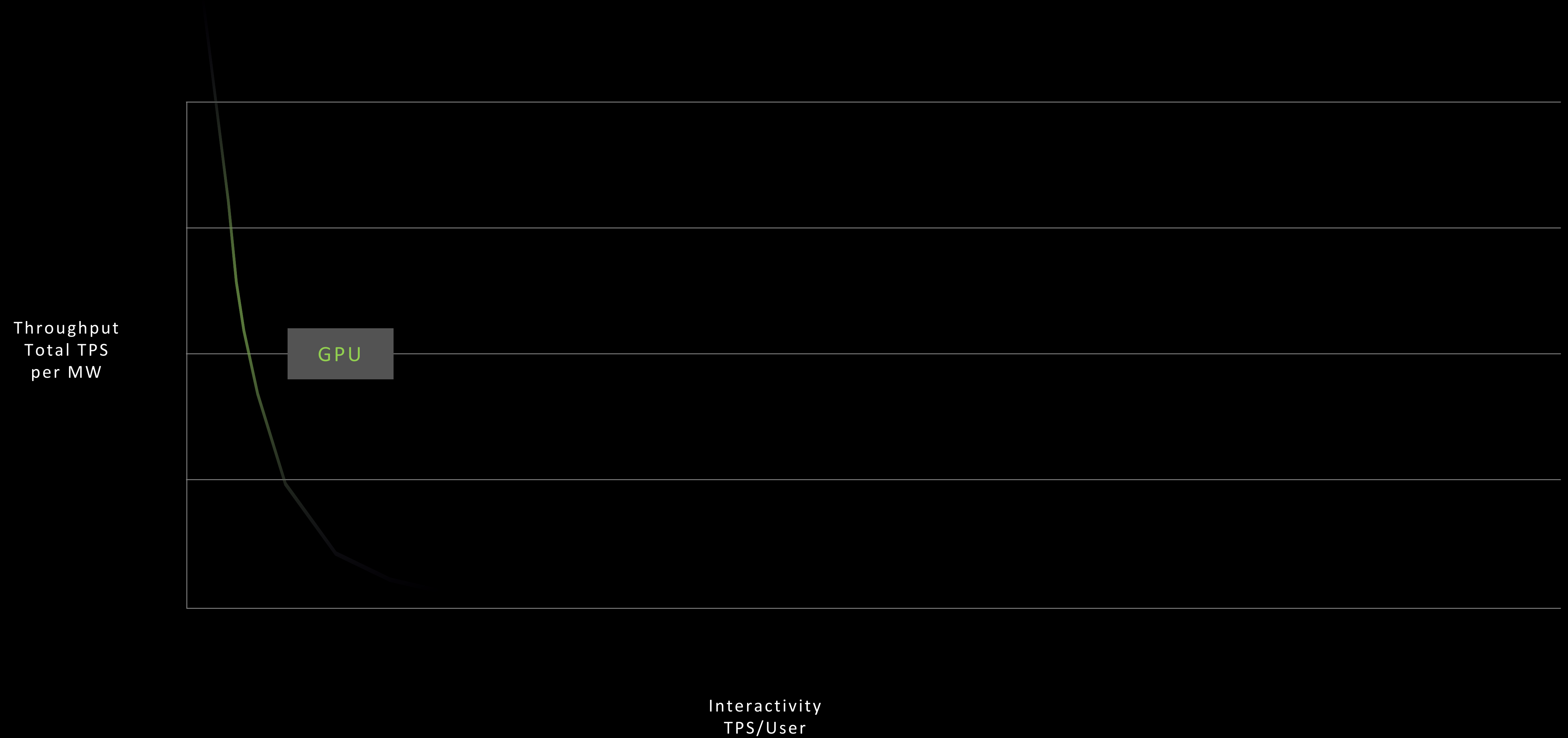
Transfer only activations
between wafers, requiring
relatively lower bandwidth

GPT-5.6 SOL X CS-3 CLUSTER



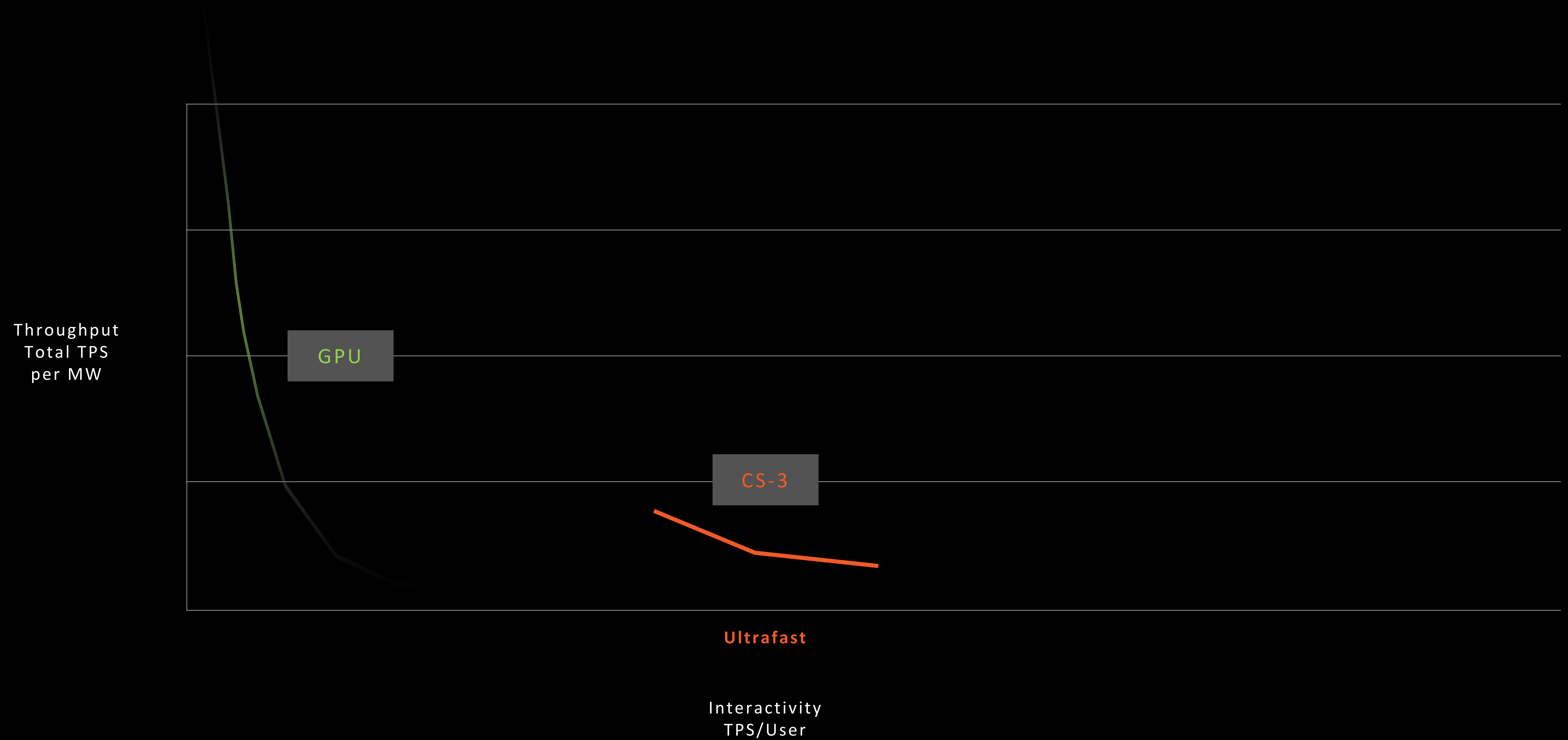
Higher I/O bandwidth means faster hops between wafers for **< 0.2 ms latency for 10T parameter model**

GPU IS REALLY SLOW AT INFERENCE



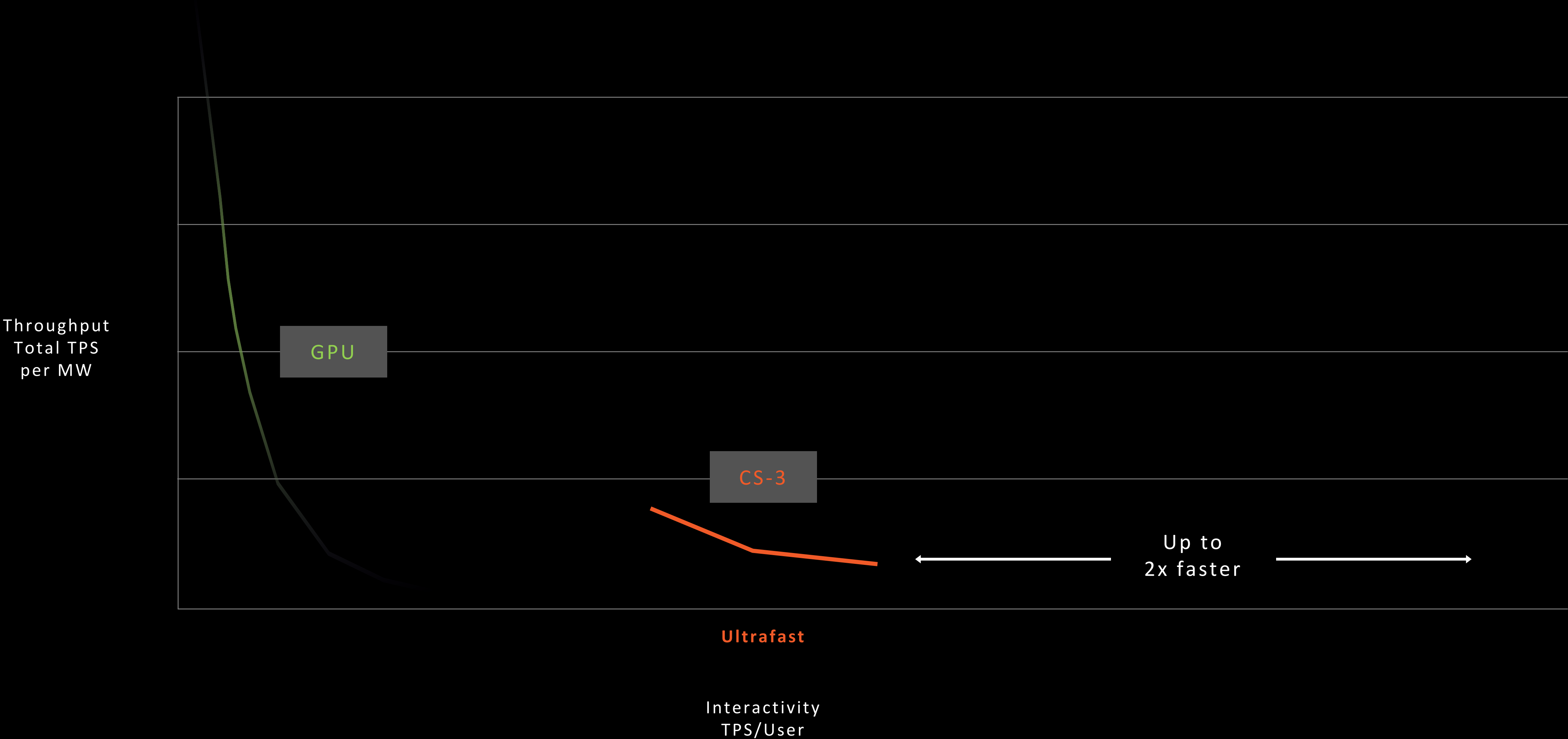
Source: Artificial Analysis and internal benchmarking (August 2026)

STARTING FROM UP TO 15x FASTER THAN GPUS



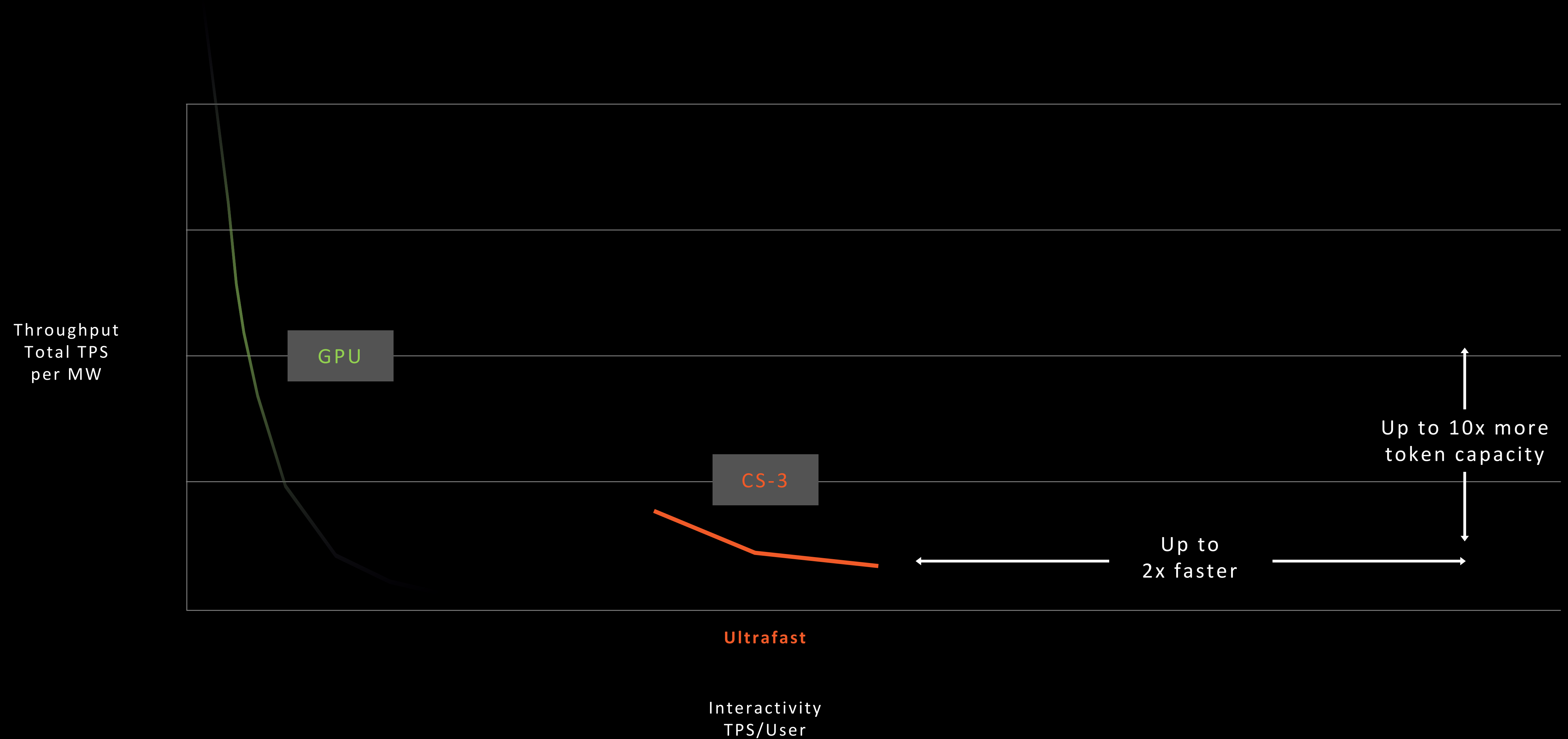
Source: Artificial Analysis and internal benchmarking (August 2026)

THE FASTEST GETS FASTER.



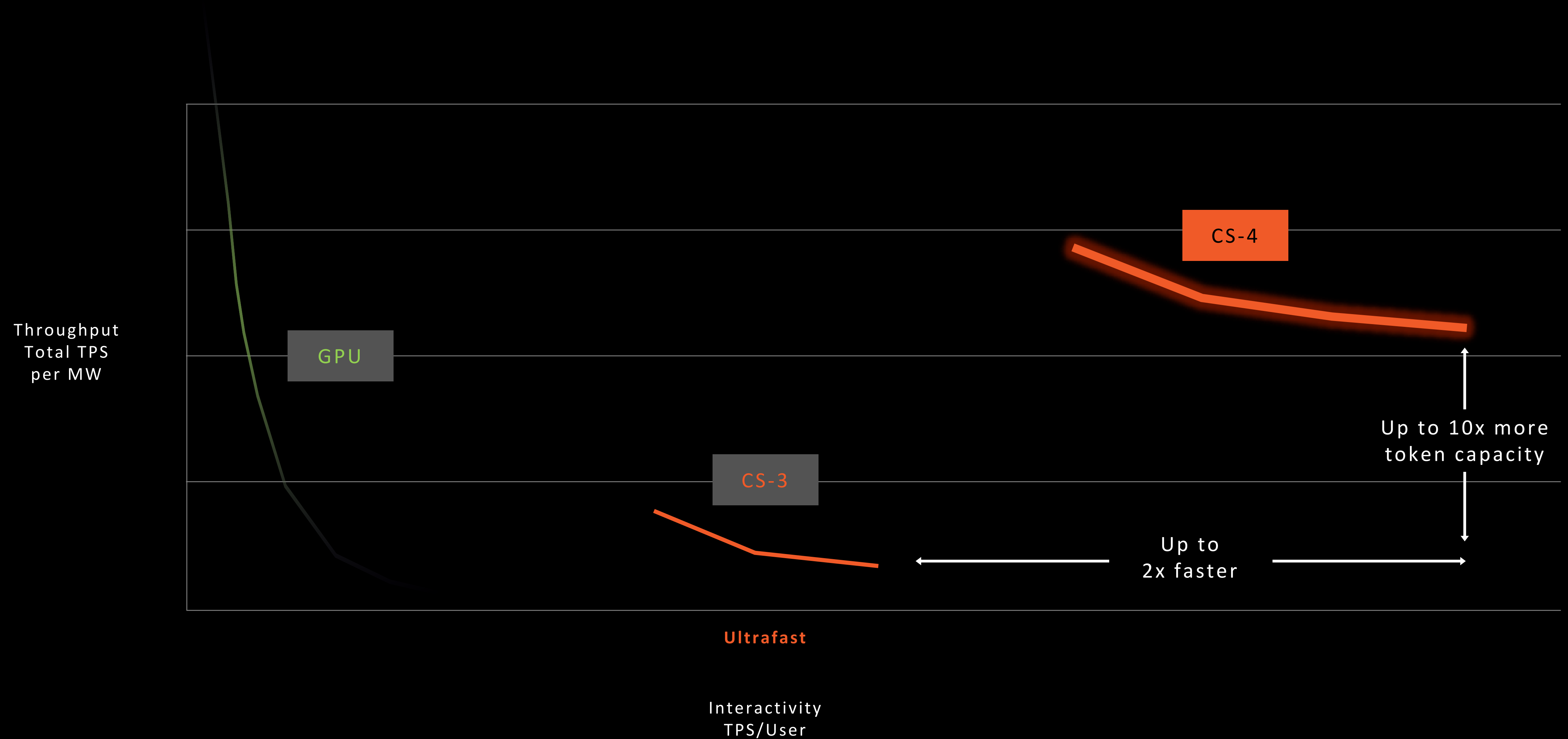
Source: Artificial Analysis and internal benchmarking (August 2026)

THE FASTEST GETS FASTER. BUILT FOR HYPERSCALE.



Source: Artificial Analysis and internal benchmarking (August 2026)

THE FASTEST GETS FASTER. BUILT FOR HYPERSCALE.



Source: Artificial Analysis and internal benchmarking (August 2026)

THE CEREBRAS CS-4 IS DESIGNED FOR
DISAGGREGATED INFERENCE

AMD

aws

...



DISAGGREGATION
IO INTERFACE



cerebras

CEREBRAS CS-4

DESIGNED FROM GROUND UP



FASTER

PERFORMANCE

Better User
Experiences

Up to 30x faster than GPUs
for more interactivity
& intelligence

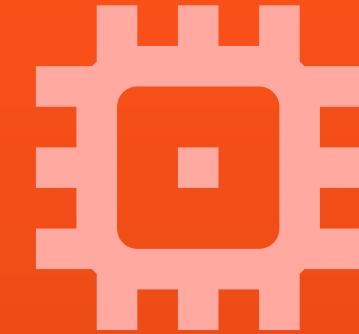


HIGHER

THROUGHPUT

Improved
Tokenomics

Up to 10x higher throughput
per watt for enhanced
tokenomics



GREATER

SCALE

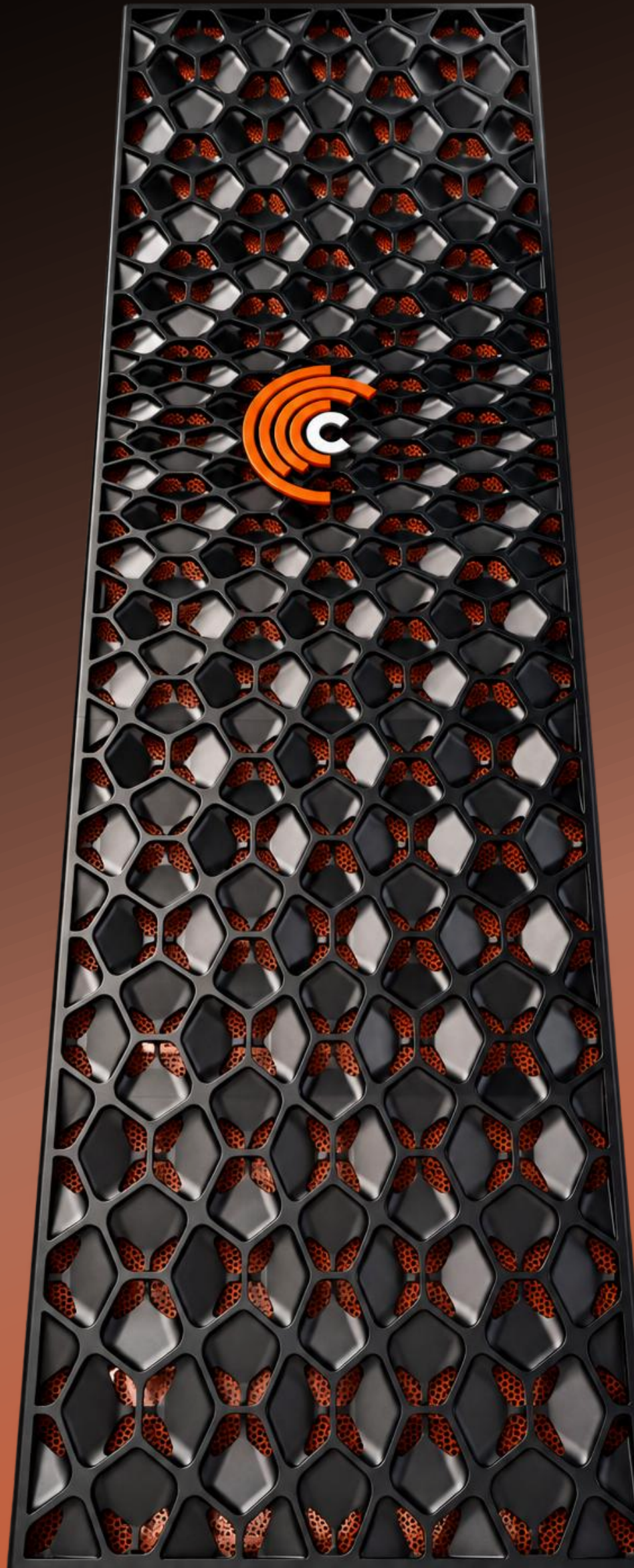
Bring More Ultrafast
Tokens to the World

Built for hyperscale with 50%
fewer components and
modular design

NEXUS RACK-SCALE PLATFORM

CS-4

EARLY ACCESS NOW, GA
LATER IN Q3 2026



CS-5 • CS-6

DESIGNED FOR MULTIPLE
GENERATIONS

CEREBRAS CS-4



SCAN THE QR CODE TO
LEARN MORE



UP TO 30X FASTER THAN
GPUS

UP TO 10X
HIGHER THROUGHPUT
PER WATT THAN CS-3

HYPERSCALE-READY
MODULAR RACK-SCALE DESIGN